SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602105

A CAPSTONE PROJECT REPORT

ON

HOUSE PRICE PREDICTION

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA1706-ARTIFICIAL IMTELLIGENCE

to the award of the degree of

BACHELOR OF TECHNOLOGY IN

COMPUTER SCIENCE ENGINEERING

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "House Price Prediction" has been carried out by G Amrutha(192511250) under the supervision of Krishnakumar Kathiresan and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, G Amrutha of the computer science engineering , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "House Price Prediction" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, datasets, software tools, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed.

Place: ____________ Date: ____________

Signature of the Students with Names

amrutha

INDIVIDUAL CONTRIBUTION STATEMENT

Student / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution (%)
Student 1	Data collection, preprocessing, model development	Dataset preparation and ML model	Model evaluation	Chapters 1, 3, 4	50%
Student 2	Feature analysis, testing, documentation	Feature engineering and interface	Performance comparison	Chapters 2, 4, 5	50%

ABSTRACT

House price prediction is a practical machine-learning problem in which the estimated value of a property depends on several interacting attributes such as location, area, number of bedrooms, bathrooms, age of the property, parking availability, and other property characteristics. Traditional valuation methods may depend heavily on expert judgment and may be difficult to scale when large numbers of properties must be evaluated. This project develops a data-driven house price prediction system that learns relationships between property features and historical sale prices.

The proposed system follows a complete machine-learning workflow consisting of dataset acquisition, data cleaning, exploratory data analysis, feature preprocessing, feature selection/engineering, training of regression models, evaluation, and prediction. Candidate algorithms include Linear Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting Regression. The models are compared using regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score. Cross-validation and a separate test set are used to reduce the risk of evaluating a model only on training data.

The expected engineering outcome is a prototype prediction model that accepts property attributes and produces an estimated house price. The system is intended as a decision-support tool rather than a replacement for professional valuation. Particular attention is given to data quality, model interpretability, fairness, privacy, security, and responsible use because property prices can affect important financial decisions.

Keywords: House Price Prediction, Machine Learning, Regression, Feature Engineering, Random Forest, Real Estate Analytics

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit

ML	Machine Learning	—
MAE	Mean Absolute Error	Price units
MSE	Mean Squared Error	$Price^2$
RMSE	Root Mean Squared Error	Price units
R^2	Coefficient of Determination	—
EDA	Exploratory Data Analysis	—

CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Real-estate pricing is influenced by multiple physical, geographic, economic, and market-related factors. A house with a larger built-up area may have a higher price, but location, neighborhood characteristics, accessibility, property age, amenities, and local demand can alter that relationship. This makes house-price estimation a suitable application of supervised machine learning, where historical property records are used to learn a mapping from input features to a continuous target value.

1.2 Need for the Project

- Property buyers need an objective reference for comparing asking prices.
- Sellers and agents need scalable estimates for preliminary valuation.
- Large property datasets make manual analysis slow and inconsistent.
- Machine-learning models can capture nonlinear relationships among multiple features.
- A transparent evaluation process can help users understand prediction uncertainty and model limitations.

1.3 Problem Statement

Develop a machine-learning-based regression system that predicts the selling price of a residential property from relevant property attributes while maintaining acceptable prediction accuracy, computational efficiency, interpretability, data quality, and responsible handling of user and property information.

1.4 Project Aim

To design and implement a reliable house price prediction prototype using supervised machine-learning regression techniques.

1.5 Project Objectives

- Collect or select a suitable historical house-price dataset.
- Clean missing, inconsistent, duplicate, and anomalous records.
- Perform exploratory data analysis and identify influential features.
- Preprocess numerical and categorical variables appropriately.
- Train and compare multiple regression algorithms.
- Evaluate models using MAE, RMSE and R^2.
- Develop a simple prediction workflow for new property inputs.
- Document ethical, security, sustainability, and risk considerations.

1.6 Scope

Included: historical residential property data, preprocessing, exploratory analysis, regression modelling, evaluation, comparison, and prototype prediction. Excluded: legal property valuation, mortgage approval, guaranteed market pricing, autonomous buying/selling decisions, and predictions for properties outside the characteristics represented in the training data. The project assumes that historical data is sufficiently representative and that the target price is recorded consistently.

1.7 Expected Engineering Outcome

The expected outcome is a tested machine-learning regression prototype with a reproducible preprocessing and prediction pipeline, model-comparison evidence, documented limitations, and a clear procedure for estimating the price of a new property.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant work was reviewed around machine-learning regression, real-estate price estimation, feature engineering, model evaluation, and responsible use of predictive systems. The review focuses on commonly used regression approaches and the engineering requirements of a practical prediction pipeline.

2.2 Review of Existing Work

Linear regression provides a simple and interpretable baseline but assumes an approximately linear relationship between predictors and price. Decision trees can model nonlinear relationships and interactions without requiring linear assumptions, although a single tree may overfit. Random forests reduce variance by averaging many decision trees and are often effective on tabular data. Gradient boosting builds an ensemble sequentially and can achieve strong predictive performance, but it requires careful parameter tuning.

Real-estate prediction systems commonly combine location and property attributes with preprocessing and validation. The main engineering challenge is not only selecting a high-performing model but also ensuring that data leakage, outliers, missing values, inconsistent units, and geographic bias do not produce misleading results.

2.3 Comparative Analysis

Existing Approach	**Advantages**	**Limitations**	**Relevance to Proposed Work**
Manual/expert valuation	Domain knowledge; contextual judgment	Slow; subjective; difficult to scale	Used as conceptual baseline
Linear Regression	Simple; interpretable; fast	Limited nonlinear modelling	Baseline model
Decision Tree	Handles nonlinear relationships	Can overfit	Candidate model
Random Forest	Robust tabular prediction; captures interactions	Less interpretable; larger model	Primary candidate
Gradient Boosting	Strong predictive accuracy	More tuning required	Candidate for comparison

2.4 Research / Engineering Gap

A practical student-level system must connect data preparation, model selection, evaluation, responsible use, and deployment into one reproducible workflow. A model with high training accuracy alone is insufficient; the project therefore emphasizes independent testing, leakage prevention, transparent metrics, and explicit limitations.

2.5 Proposed Contribution

The proposed work contributes an end-to-end house price prediction pipeline in which multiple regression models are evaluated under the same preprocessing and test conditions. The selected model is based on measured validation performance rather than preference, while ethical, security, and risk considerations are included in the engineering design.

CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Buyer	Receive a quick indicative estimate
Seller/Agent	Compare properties consistently
Project Evaluator	Reproducible model and metrics
Developer	Maintainable preprocessing and prediction pipeline

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Accept property features as input	Functional	Valid input produces prediction
Preprocess missing/categorical data	Functional	Same pipeline used in training and prediction
Train multiple regressors	Functional	At least 3 models compared
Evaluate with MAE/RMSE/R^2	Functional	Metrics reported on unseen data
Response time	Non-Functional	Prediction produced within a few seconds on project hardware
Reproducibility	Non-Functional	Fixed preprocessing and documented parameters
Security	Non-Functional	No unnecessary personal data stored

3.2 Constraints & Applicable Standards

Applicable constraints include dataset licensing, privacy, data quality, computing resources, model bias, and responsible communication of predictions. Where software and data are obtained externally, their respective licenses and terms of use must be respected. Personal information should not be collected unless necessary for the stated project purpose.

Standard / Code	Requirement	How Applied in This Project
IEEE 7000-2021	Model ethical considerations in system design	Ethical risks and stakeholder impact considered
ISO/IEC 25010	Software quality characteristics	Reliability, usability, performance and maintainability considered
Dataset/software license	Permitted use and attribution	Dataset and libraries acknowledged in references

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Prediction metric	Minimize RMSE	Price units	High
R^2	Higher is better	—	High
Input validation	Reject invalid/missing required values	—	High
Prediction latency	Few seconds or less	s	Medium
Reproducibility	Fixed pipeline and documented split	—	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Linear Regression is used as a transparent baseline and provides an easy-to-understand reference.

Alternative 2: Random Forest Regression models nonlinear relationships and interactions using an ensemble of decision trees.

Alternative 3: Gradient Boosting Regression can provide strong tabular-data performance but may require more tuning.

Criterion	Weight	Linear Regression	Random Forest	Gradient Boosting
Performance	40	Medium	High	High
Cost/complexity	20	High	Medium	Medium
Interpretability	15	High	Medium	Medium
Robustness	15	Medium	High	High
Maintainability	10	High	High	Medium

3.5 Selected Design & Detailed Design

Random Forest or the best-performing validated model is selected based on the evaluation results in Chapter 4. The final pipeline consists of input validation, preprocessing, feature transformation, trained regression model, prediction, and result presentation.

System architecture: User Input → Validation → Preprocessing → Trained Regression Model → Predicted Price → Result / Interpretation.

The pipeline must apply identical preprocessing rules to training and new prediction data. This prevents mismatched feature encoding and scaling. Training data is separated from the test set before model evaluation so that test information does not influence training.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Model complexity	Linear regression	Simple and interpretable	May miss nonlinear effects	Use as baseline; select final model by test metrics
Ensemble model	Random forest vs boosting	Better nonlinear modelling	Higher complexity	Prefer validated best performer
Feature set	All available vs selected	More information vs noise	Too many weak features can hurt generalization	Use evidence-based features

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational efficiency	Training time and model size	Avoid unnecessarily complex models
Ethics	Fairness and explainability	Feature importance, subgroup errors where data permits	Do not present prediction as guaranteed value
Health & Safety	Financial decision risk	Potential impact of inaccurate estimates	Clear disclaimer and human review
Security	Data protection	Inputs, storage, access and output exposure	Avoid personal data and secure stored data

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Poor data quality	Medium	High	High	Cleaning, validation, EDA	Medium
Overfitting	Medium	High	High	Cross-validation, test set	Low-Medium
Data leakage	Low-Medium	High	High	Split before preprocessing; pipeline	Low
Bias in historical data	Medium	High	High	Audit features/errors; document limits	Medium
Unexpected input	Medium	Medium	Medium	Input validation	Low

CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows an iterative machine-learning workflow: define the target, inspect the dataset, clean and preprocess the data, explore feature relationships, create training and testing partitions, train candidate models, validate and compare performance, then package the selected model into a prediction workflow. Errors identified during evaluation are used to refine preprocessing and model selection.

Resource	Purpose
Python	Programming and experimentation
Pandas / NumPy	Data manipulation
Scikit-learn	Preprocessing, regression and evaluation
Matplotlib	Plots and model analysis
Jupyter Notebook / IDE	Development and documentation
House-price dataset	Training and evaluation data

4.2 Software Implementation & Modern Tools Used

Tool / Software	Purpose	Stage Used
Python	Core implementation	All stages
Pandas	Cleaning and analysis	Data preparation
NumPy	Numerical operations	Preprocessing/modeling
Scikit-learn	Models and metrics	Training/evaluation
Matplotlib	Visualization	EDA/results
Git (if used)	Version control	Development

Implementation evidence to insert: dataset preview, preprocessing output, model-training screenshot, actual-vs-predicted plot, and final prediction interface/output.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Input validation	Enter valid and invalid property values	Invalid values rejected or handled safely

Preprocessing	Test missing/categorical inputs	Consistent transformed feature representation
Model training	Train candidate regressors	Models train without leakage/errors
Prediction	Use unseen test records	Predictions generated successfully
Accuracy	Calculate MAE/RMSE/R^2	Metrics reported on held-out data

Specification	**Required Value**	**Achieved Value**	**Status**
Training/test separation	Separate unseen test set	To be measured after execution	Pending
Regression metrics	MAE, RMSE, R^2	To be measured	Pending
Prediction workflow	Valid new input produces estimate	To be measured	Pending

4.4 Results

Because the report template does not contain a supplied experimental dataset or measured model outputs, numerical performance values are not fabricated here. After implementation, this section should contain the actual MAE, RMSE and R^2 for every candidate model, together with plots of residuals and actual-versus-predicted prices.

Model	**MAE**	**RMSE**	**R^2**	**Rank**
Linear Regression	To be measured	To be measured	To be measured	—
Decision Tree Regression	To be measured	To be measured	To be measured	—
Random Forest Regression	To be measured	To be measured	To be measured	—
Gradient Boosting Regression	To be measured	To be measured	To be measured	—

4.5 Data Analysis & Engineering Judgment

The final model should be judged primarily using unseen-test performance and cross-validation rather than training accuracy. RMSE is useful when larger prediction errors should receive greater penalty, while MAE provides an easier-to-interpret average absolute error. R^2 indicates the proportion of target variance explained by the model under the evaluation data. If training performance is much better than test performance, overfitting should be investigated.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	**Evidence**	**Achievement**
Collect/select dataset	Dataset documentation	Fully after dataset is finalized
Clean and preprocess	Preprocessing pipeline	To be verified
Train multiple models	Model comparison table	To be verified
Evaluate performance	MAE/RMSE/R^2 on test set	To be verified
Build prediction workflow	New-input prediction	To be verified
Document responsible use	Chapter 3 risk/ethics analysis	Fully documented

4.7 Strengths & Limitations

Strengths:

- End-to-end machine-learning workflow.
- Comparison of multiple regression algorithms.
- Use of objective regression metrics.
- Explicit treatment of data leakage, ethics, risk, and security.

- Can be extended with new data and additional features.

Limitations:

- Predictions depend on the quality and representativeness of historical data.
- Market conditions can change after model training.
- Location and neighborhood variables may be difficult to encode comprehensively.
- Historical prices can contain bias that the model may reproduce.
- The system is an indicative prediction tool, not a certified property valuation.

4.8 Project Planning, Roles & Budget

Milestone	**Weeks**	**Output**
Problem definition and literature review	1–2	Requirements and review
Dataset preparation	3–4	Clean dataset
EDA and feature engineering	5–6	Feature set and plots
Model development	7–8	Candidate regressors
Testing and evaluation	9–10	Metrics and comparison
Documentation and presentation	11–12	Final report and demo
Student	**Assigned Responsibility**	**Actual Contribution**
Student 1	Data preparation and model development	To be updated
Student 2	Evaluation, documentation and presentation	To be updated
Item	**Estimated Cost**	**Actual Cost**
Software libraries	₹0 (open source)	₹0
Computing	Existing system	To be updated
Dataset	Depends on source/license	To be updated
Documentation/printing	As required	To be updated

CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The House Price Prediction project presents an engineering approach to estimating residential property prices using supervised machine learning. The system integrates data preparation, exploratory analysis, feature processing, regression modelling, testing, and responsible-use considerations into a single workflow. Multiple models are compared using objective regression metrics, with the final model selected from measured validation and test performance. The resulting prototype can provide fast indicative estimates while clearly acknowledging that predictions depend on the quality, coverage, and currency of the training data.

5.2 Future Work

- Integrate richer geographic and neighborhood features.
- Add time-dependent market indicators to handle changing price trends.
- Use hyperparameter optimization and ensemble stacking where justified.
- Provide prediction intervals or uncertainty estimates.
- Deploy the model as a web/mobile interface with secure APIs.
- Monitor model drift and retrain using validated new data.
- Evaluate fairness and error patterns across relevant geographic groups.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired: machine-learning regression, data preprocessing, model validation, feature engineering, evaluation metrics, and responsible AI considerations.

Engineering & Professional Skills Developed: problem formulation, data analysis, Python implementation, experimental design, technical documentation, teamwork, and evidence-based decision making.

Individual Reflection – Student 1: The main challenge was preparing property data so that different feature types could be handled consistently. The solution was to build a repeatable preprocessing workflow and validate it before model comparison. A key engineering decision was to compare multiple models rather than assuming one algorithm would always be best. If the project were repeated, greater emphasis would be placed on geographic and time-based validation.

Individual Reflection – Student 2: The major learning outcome was understanding how model metrics translate into engineering decisions. Comparing MAE, RMSE, and R^2 showed why a single accuracy measure is not sufficient for a regression system. The project also highlighted the importance of responsible communication because a price estimate can influence financial decisions. A future version would include uncertainty estimates and stronger monitoring.

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[6] IEEE Std 7000-2021, IEEE Standard Model Process for Addressing Ethical Concerns During System Design, 2021.

[7] Scikit-learn documentation, Machine Learning in Python. Accessed for implementation guidance.

[8] Pandas documentation, Python Data Analysis Library. Accessed for data preparation guidance.

APPENDICES

Appendix A – Detailed Calculations

Include metric formulas and any detailed preprocessing or feature-engineering calculations used in the final experiment.

Appendix B – Engineering Drawings / Schematics

Insert the final system architecture and machine-learning workflow diagram.

Appendix C – Source Code / Code Repository Information

Include essential code excerpts or the approved repository information. The complete source code may be maintained separately.

Appendix D – Datasheets

Not applicable to a software-only implementation unless external computing hardware is specifically documented.

Appendix E – Experimental / Raw Data

Include dataset description, train/test split information, model outputs, and final evaluation tables.

Appendix F – Ethics / Institutional Approval

Include applicable approval or declaration documentation, if required.

Appendix G – Project Meeting / Progress Records

Include meeting notes and milestone records.

Appendix H – User/Industry Feedback

Include feedback if collected.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Include only if applicable.

Appendix J – Individual Contribution Evidence

Include evidence supporting team-member contributions.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)